

Felipe Arbulú

POSTDOCTORAL RESEARCHER

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Professional Experience

2025-2026 **Postdoctoral researcher**, LAMFA, Université de Picardie Jules Verne
2023-2025 **Attaché Temporaire d'Enseignement et de Recherche**, LAMFA, Université de Picardie Jules Verne

Education

Université de Picardie Jules Verne *LAMFA, Amiens, France*
PHD IN MATHEMATICS *2020 - 2024*

- Thesis: Spectral analysis of finite topological rank $\mathcal{S}$ -adic subshifts
- Advisor: Fabien Durand

Université Paris-Saclay *IMO, Orsay, France*
MS IN MATHEMATICS *2019 - 2020*

- Thesis: Periodicity and ergodicity in the trihexagonal tiling
- Advisor: Samuel Lelièvre

Universidad de Chile *FCFM, Santiago, Chile*
MS IN ENGINEERING MENTION MATHEMATICS *2017 - 2018*
Universidad de Chile *FCFM, Santiago, Chile*
CIVIL ENGINEERING MENTION MATHEMATICS *2012 - 2018*

- Thesis: Contributions to the study of eigenvalues and weak mixing for interval exchange transformations and affine geometric systems
- Advisor: Alejandro Maass

Universidad de Chile *FCFM, Santiago, Chile*
BS IN ENGINEERING MENTION MATHEMATICS *2012 - 2015*

Publications

PUBLISHED

The Jacobs–Keane theorem from the $\mathcal{S}$ -adic viewpoint. Arbulú, Felipe, Durand, Fabien and Espinoza, Bastián. 2024. Discrete Contin. Dyn. Syst. pp. 3077–3108. doi: 10.3934/dcds.2024052 • arXiv: 2307.10663.
Dynamical properties of minimal Ferenczi subshifts. Arbulú, Felipe and Durand, Fabien. 2023. Ergodic Theory Dynam. Systems. pp. 3923–3970. doi: 10.1017/etds.2023.7 • arXiv: 2207.14097.

PREPRINTS

Topological weak mixing for certain linear involutions of hyperelliptic type. Arbulú, Felipe, Espinoza, Bastián and Maass, Alejandro. 2026. arXiv: 2607.16830.

Awards, Honors and Grants

- 2020 **Becas Chile (scholarship for PhD. studies abroad)**, Chilean Government
- 2019 **Master 2 scholarship**, Fondation Mathématique Jacques Hadamard, France
- 2017 **Beca de Excelencia CMM (graduate scholarship)**, Center for Mathematical Modeling, Chile
- 2016 - 2017 **Beca Bicentenario (undergraduate scholarship)**, Ministry of Education, Chile
- 2016 - 2017 **Summer Scholarship**, Instituto de Matemática Pura e Aplicada, Brazil
- 2013 - 2017 **Outstanding Student**, FCFM, Universidad de Chile, Chile
- 2011 **Participation**, International Mathematical Olympiad, Netherlands
- 2010 **Honorable Mention**, Iberoamerican Mathematical Olympiad, Paraguay
- 2010 **Participation**, Cono Sur Mathematical Olympiad, Brazil
- 2007 - 2011 **Medalist**, National Mathematical Olympiad, Chile

Talks

CONFERENCES

- June 4, 2026. **Topological weak mixing for certain linear involutions of hyperelliptic type.**
Dyadisc 9: Expensive Dynamics – The dynamic youth speaks out, Amiens, France.
- June 30, 2024. **The Jacobs–Keane theorem from the $\mathcal{S}$ -adic viewpoint.**
Workshop ANR IZES, Île de Porquerolles, France.
- May 13, 2024. **The Jacobs–Keane theorem from the $\mathcal{S}$ -adic viewpoint.**
Dynamics on zero-dimensional spaces: new connections, Leiden, Netherlands.
- July 7, 2023. **Dynamical properties of minimal Ferenczi subshifts.**
Dyadisc 6. Dynamics on the Cantor set and applications, Amiens, France.
- June 17, 2022. **Dynamical properties of minimal Ferenczi subshifts.**
Dyadisc 5. $\mathcal{S}$ -adicity, Liège, Belgium.
- November 29, 2021. **Dynamical properties of symbolic rank one subshifts.**
SDA2: Systèmes Dynamiques, Automates et Algorithmes, Caen, France.
- July 8, 2021. **Online Poster: Rank one subshifts.**
Dyadisc4. Problèmes ouverts en dynamique de faible complexité, Amiens, France.

SEMINARS

- November 6, 2025. **Analyse spectrale de certaines classes de sous-shifts $\mathcal{S}$ -adiques.**
Séminaire de Théorie Spectrale et Géométrie, Grenoble, France.
- April 11, 2025. **Spectral analysis of some classes of $\mathcal{S}$ -adic subshifts.**
The Budapest-Wien Dynamics Seminar, Budapest, Hungary.
- May 31, 2023. **Dynamical properties of minimal Ferenczi subshifts.**
Séminaire de Mathématiques discrètes, Liège, Belgium.
- February 2, 2023. **Dynamical properties of minimal Ferenczi subshifts.**
Séminaire équipe ADA, Calais, France.
- January 10, 2023. **Dynamical properties of minimal Ferenczi subshifts.**
Séminaire Ernest, Marseille, France.
- March 30, 2021. **Dynamical properties of symbolic rank one subshifts.**
Séminaire Dynamique et Probabilités, Amiens, France.

Teaching

Spring 2026	L1S2: Analyse réelle fondamentale , Teaching Assistant	<i>UPJV, France</i>
Fall 2026	L1S1: Méthodes et Techniques de Calcul , Teaching Assistant	<i>UPJV, France</i>
Spring 2025	M1S2: Systèmes dynamiques , Teaching Assistant	<i>UPJV, France</i>
Spring 2025	M1S2: Modélisation aléatoire , Teaching Assistant	<i>UPJV, France</i>
Spring 2025	L1S2: Analyse réelle appliquée , Teaching Assistant	<i>UPJV, France</i>
Spring 2025	L1S2: Analyse réelle fondamentale , Teaching Assistant	<i>UPJV, France</i>
Spring 2025	L1S2: Probabilités et statistique , Teaching Assistant	<i>UPJV, France</i>
Fall 2025	L1S1: Méthodes et Techniques de Calcul , Teaching Assistant	<i>UPJV, France</i>
Spring 2024	M1S2: Systèmes dynamiques , Teaching Assistant	<i>UPJV, France</i>
Spring 2024	M1S2: Modélisation aléatoire , Teaching Assistant	<i>UPJV, France</i>
Spring 2024	L1S2: Analyse réelle appliquée , Teaching Assistant	<i>UPJV, France</i>
Spring 2024	L1S2: Analyse réelle fondamentale , Teaching Assistant	<i>UPJV, France</i>
Spring 2024	L1S2: Probabilités et statistique , Teaching Assistant	<i>UPJV, France</i>
Fall 2022	L1S1: Méthodes et Techniques de Calcul , Teaching Assistant	<i>UPJV, France</i>
Fall 2018	MA5112: Differential and Riemannian Geometry , Teaching Assistant	<i>FCFM, Chile</i>
Fall 2017	MA5112: Differential and Riemannian Geometry , Teaching Assistant	<i>FCFM, Chile</i>

Other activities

PEER REVIEW

Journals. Theoretical Computer Science.

MATH OLYMPIAD TRAINER

Trainer of the Chilean team for international mathematical competitions between 2020 and 2023.

Skills

PROGRAMMING

• Python • C++ • SageMath • $\text{\LaTeX}$ • MATLAB • Git

LANGUAGES

• Spanish: First language
• English: Fluent
• French: Proficient